

MECH 3620 – Aircraft Design

Spring 2025–2026

Final Project

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1 Description

(Note that the texts in [dark blue](#) contain hyperlink – so hover and explore.)

Your group has recently established itself as an aircraft manufacturing company specializing in high-efficiency regional mobility solutions. You have been approached by Ms. Y from the Horizon Group, a consultancy firm representing a consortium of regional airlines in the Greater Bay Area and Southeast Asia. She has requested your company to design a conceptual Regional Jet (RJ) tailored for high-frequency, short-to-medium haul operations.

The proposed aircraft is expected to serve as a regional backbone, linking secondary cities to major hubs and facilitating direct point-to-point services on routes where larger narrow-body jets are economically inefficient. Operating primarily from [Hong Kong International Airport \(HKG\)](#), the aircraft is optimized for high-frequency sectors: [Da Nang International Airport \(DAD, ~ 930 km\)](#), [Wuhan Tianhe International Airport \(WUH, ~ 940 km\)](#), and [Clark International Airport \(CRK, ~ 1050 km\)](#). The design philosophy prioritizes operational flexibility, rapid turnaround times, and superior economic efficiency (low cost per trip), all while ensuring a competitive passenger experience.

Below, you can find the specific requirements and expectations for the project.

1. General requirements

- (a) The aircraft must be suitable for regional commercial operations (Small Regional Jet).
- (b) Passenger capacity must range from 50 to 70 passengers in a two-class configuration, with the assumed design weights of 90 kg per passenger plus 15 kg of luggage.
- (c) Maximum takeoff weight (MTOW) **must not exceed** 35 000 kg.
- (d) The aircraft must be compatible with [ICAO Aerodrome Reference Code C](#) standards. The wingspan **must not exceed** 28 m to ensure access to constrained ramps (Note: Code B wingspans (<24 m) are acceptable and encouraged if performance targets are met).
- (e) Cargo capacity must accommodate **at least** 12 m³ of baggage and freight.

2. Performance requirements

- (a) **Design Range:** To minimize turnaround times and ground support dependency at outstations, the aircraft must be capable of performing a round-trip mission between HKG and the designated airports **without refueling** at the destination. The aircraft must carry sufficient fuel for both trips while transporting a **maximum payload**, retaining [standard ICAO fuel reserves](#) (alternate + holding + contingency) upon the final return to HKG.
- (b) **Service Ceiling:** The aircraft must have a maximum service ceiling of **at least** 12 497 m (41 000 ft) to enable vertical avoidance of severe tropical convective weather systems com-

mon in Southeast Asia and to secure access to uncongested flight levels above the high-density traffic of the Pearl River Delta region.

- (c) **Takeoff:** Certified takeoff field length (TOFL) at MTOW **must not exceed** 1800 m (ICAO Aerodrome Code Number 3) at sea level in ISA+15°C conditions.
- (d) **Landing:** Certified landing distance with maximum payload and necessary reserves **must not exceed** 1600 m at sea level in ISA+15°C conditions.
- (e) **Climb:** Minimum Time to Climb (TTC) to an initial cruise altitude of 10 668 m (35 000 ft) at MTOW **must not exceed** 18 min, needed by the obstacle clearance requirements of complex terrain (e.g., Lantau Peak) and the highly congested airspace surrounding HKG.
- (f) **Cruise:** The aircraft must achieve a cruise speed of **at least** Mach 0.74.

3. Certification requirement

- (a) The aircraft must be certifiable under FAR-25 regulations or their equivalent.

4. Environmental and operational requirements

- (a) The design must demonstrate a reduction in fuel consumption per seat-kilometer compared to current in-service regional jets (e.g., ERJ Family, CRJ Series).
- (b) Noise levels must comply with ICAO Chapter 14 noise standards or better.
- (c) CO₂ emissions per seat must be reduced compared to existing regional jets.
- (d) The aircraft must support a turnaround time of **30 minutes or less**, based on a mission profile that does not require refueling at the outstation (fuel tankering). To facilitate this within the constraints of Code C ramp spacing, the design **must allow** for:
 - Single-point pressure refueling.
 - Simultaneous passenger boarding and cargo loading (separation of service doors).
 - Compatibility with standard ground handling equipment without specialized adapters.

You need to be aware that there are several competitors in this market. The more your design exceeds their performance, the higher your chances will be to win this tender (i.e., to outperform competitors in the business world and achieve a higher grade in the academic setting).

At the end of the project, you will need to prepare a technical report that contains the following information:

1. The detailed mission profile formulation based on the customer's requirements.
2. The aircraft configuration (including the aerodynamic design) that best fits this mission.
3. The performance evaluation of your aircraft and justification that it can meet the requirements – you can supplement it with figures, charts, and diagrams as appropriate.

In addition to the aforementioned main requirements, you should also consider additional *Measures of Merit (MoM)* that will make your design more attractive than those of your competitors. These could include, but are not limited to:

- Advanced avionics or safety systems that enhance operational reliability.
- Features that improve passenger comfort, such as wider seats or advanced cabin air systems.
- Use of innovative materials (e.g., composites) or compatibility with alternative fuels (e.g., SAF or hydrogen).
- Future-proofing for future variants, such as hybrid-electric or hydrogen propulsion systems.
- Improvements in cost efficiency, balancing production costs and maintenance costs.

Note that this project focuses on the **conceptual design stage**, so your design goals must be realistic. Innovative ideas are encouraged, and they may give Ms. Y a surprise and make her favor your design over those of your competitors. However, they need to be properly backed up with literature review and preliminary investigations to convince your customer.

This project is designed to challenge your understanding of aircraft design and the compromises involved in meeting conflicting requirements. To succeed, **you will need to:**

1. Balance trade-offs between range, payload, and fuel efficiency within the constraints of weight, size, and performance.
2. Justify your design decisions and demonstrate how they meet the specified requirements.
3. Propose creative and innovative solutions, supported by technical feasibility studies.

2 Guidelines to Complete the Project

The AeroFuse platform has been specifically designed to help you with this project. All the basic functionalities will be provided and explained during tutorial sessions, but feel free to go beyond what's asked. The **four milestones** listed below provide a guideline for you to complete your project. **Hints** are generously provided to help you with some tasks. **Sanity checks** are given to help you ensure that you are on the right track, i.e., preventing you from designing an aircraft that is unrealistic or not compliant to Ms. Y's requirements. While some code examples/templates will be given throughout the semester, you will have the opportunity to hone your coding skills by writing some of them yourselves.

To ensure that you can finish your project on time, complete each milestone on or before the suggested completion date. Everything that you generate to complete the milestone tasks will need to be included in the final report, so make sure you read the descriptions carefully. Please approach the TAs when you have any difficulties completing your milestones.

2.1 Use of AeroFuse

AeroFuse is a computational platform which will help you perform analyses of aircraft designs. It currently features tools for analyzing the geometry, aerodynamics, and structures of aircraft surfaces. It aims to introduce elementary aspects of programming alongside instructive demonstrations of aircraft design and analysis. In the tutorial sessions during the semester, some introductory notebooks will be provided with the tutorials for you to develop familiarity with the platform.

2.2 Milestone 1: Market Analyses & Mission Requirements

Suggested completion date: February 23, 2026.

Hint: This section does not require you to use any computational tools. You will need to provide your business insight.

You will start this project by deciding on the specific mission requirements for your aircraft, based on the main requirements stated above and your intended MoM's (if any). You are advised to perform a market research to decide on a particular route that your aircraft will be optimized for, which should be aligned with the need of your potential customer. These mission requirements will determine the required performance parameters of your aircraft later on.

To start with, fill out the comparison table as shown in Table 1 (you can include as many aircraft as you want – it is always better to have more reference data). The information you put here will be useful as your reference when doing *sanity check* throughout the project phases.

Hint: You should start by finding some existing aircraft types under regional jet airliners.

From your market analyses and the comparison you performed just now, extract a list of requirements that your aircraft needs to satisfy to meet the customer's requirements. This should include things

Table 1: Comparison tables of similar aircraft currently available on the market.

Parameter	Aircraft #1	Aircraft #2	Aircraft #3	...
Model Name				
No of Passengers				
Max Range				
MTOW				
Payload				
Fuel Weight				
Cruise Speed				
Cruise Altitude				
Service Ceiling				
Time to Climb				
Engine				
Thrust				

such as: number of crews and passengers, payload, take-off and landing conditions, mission profile and parameters, and other functional requirements that you have identified in your MoM. The aircraft should also satisfy [Part 25 of the Federal Aviation Regulations \(FAR-25\)](#). Start with filling out Table 2, and then add more information (e.g., number of crews) as you see fit. Make sure you have enough information to enable you to proceed to Milestone 2.

Table 2: Overall performance parameters based on the requirements.

Number of seats		MTOW	
Cruise speed		Payload	
Cruise altitude		Engine	
Max range		Thrust	

Sanity checks: Will your proposal satisfy Ms. Y's tender document? Is it comparable or even better than your competitors? Is your design goal realistic?

2.3 Milestone 2: Preliminary Weight Estimation and Wing Design

(This includes preliminary sizing, weight estimation, configuration selection, wing design, and fuselage layout and cockpit design.)

Suggested completion date: March 13, 2026.

Hint: You have learned some aircraft conceptual design theories, some of which have been integrated into the above-mentioned **AeroFuse** platform. Some of the basic usages have been introduced during tutorial sessions and now it is an opportunity to explore the potential of this powerful tool.

For the preliminary weight estimation, please fill out Table 3 with your design, and describe how you come up with this result (including parameters and assumptions used).

Hint: Do you remember which Pluto notebook(s) can help you with initial weight estimation?

Table 3: Preliminary weight estimation

	Gross TO Weight	Empty Weight	Fuel Weight	Payload	Crew Weight
Metric Units (km, m, kg)					
Imperial Units (nmi, ft, lb)					

For preliminary sizing, please provide the **matching chart** (constraint analysis plot). You will need to show all the equations you used and assumptions you made to generate the matching chart. Please also show your **design point** on the matching chart and justify your design decision. Also fill out Table 4 with your aerodynamic parameters.

Hint: Do you remember which Pluto notebook(s) can help you with drag estimation and matching chart?

Table 4: Aerodynamic performance table (should contain the *nominal* values).

	Takeoff	Cruise	Landing
C_L			
e			
$C_{D,0}$			
L/D			

Sanity checks: Why do we need a different C_L for a different flight phase, e.g., takeoff and cruise? How does cruise C_L compare with the $C_{L_{\max}}$ of the wing?

Prepare a **preliminary three-view sketch** for your configuration and show all the basic dimensions. Fill out Table 5 with your wing design parameters and Table 6 with your fuselage design parameters. Feel free to include other diagrams that can help explain your design further. All key design considerations and features that go into the layout should be mentioned and described clearly (including assumptions and/or references, if applicable).

Table 5: Wing layout

Parameter	Value
High, mid, or low wing	
Reference area (sqm)	
Aspect ratio	
Root chord (m)	
Taper ratio	
Leading-edge sweep (deg)	
Airfoil	
Dihedral (deg)	
Incidence (deg)	

Table 6: Fuselage layout

Parameter	Fuselage	Cabin Interior
Length (m)		
Maximum height (m)		
Maximum width (m)		
S_{wet} (sqm)		
S_{frontal} (sqm)		

Based on the engine selection and wing design that you have obtained at this stage, calculate the **Time to Climb (TTC)** to FL350 (Flight Level 350; altitude of 35000 ft) of your aircraft, using a time-integration method. **Is your calculated TTC compliant to Ms. Y's requirement?** If not, you need to improve your design and re-calculate TTC, until it meets the requirement.

The outcomes from milestones 1 and 2 should be included in your **midterm project review**.

2.4 Milestone 3: Preliminary Weight & Balance, Stability Analysis, Stabilizer and Landing Gear Design

Suggested completion date: April 10, 2026.

Hint: AeroFuse also provides handy tools to size the stabilizer and landing gear. Make full use of this platform!

Fill out Table 7 with the relevant parameters for your **preliminary weight and balance analysis** (make sure you specify the units clearly). Feel free to add more components (or further break down the listed group) to the table as you see fit. You will also need to state your CG position (x_{cg}) and state the forward and aft CG limits.

Table 7: Center of gravity breakdown

Component	Weight	X-Distance
Fuselage		
Wing		
H-Tail		
V-Tail		
Landing Gear		
Crew		
Fuel		
Passenger		

Sanity checks: Is your MTOW similar to that of a regional jet airliner? Is the weight the same as that in Milestone 2? If not, what should be done?

For the **stability analysis**, please specify the neutral point location (x_{np}), CG position (x_{cg}), mean aerodynamic chord (MAC), and static margin.

Hint: Do you remember which Pluto notebook(s) can help you with tail sizing, weight and balance, and static margin estimation?

For **tail design**, you will need to fill out Table 8 and Table 9.

Table 8: Tail layout

Parameter	H-Tail	V-Tail
Reference area (sq. m)		
AR		
Chord, Root (m)		
Span (m)		
Taper Ratio		
LE Sweep (deg)		
Airfoil		
Incidence		
Volume Coefficient		

Table 9: Control surface layout

	Flaps	Slats	Aileron	Elevator	Rudder
Inboard span location (% of $b/2$)					
Outboard span location (% of $b/2$)					
Inboard chord fraction (% chord)					
Outboard chord fraction (% chord)					

Hint: If you still remember MECH 3670 – Aircraft Performance and Stability, this is where you use stability derivatives to guide the design of an aircraft.

For the **landing gear design and configuration**, you need to specify the landing gear size, locations, and integration.

2.5 Milestone 4: Structural Analyses, Propulsion System, Refinements, and Cost Analysis

Suggested completion date: April 24, 2026.

Structural Analyses Show the **V-n diagram**.

Hint: Do you remember which Pluto notebook(s) can help you with V-n diagram?

Propulsion System Describe your **engine selection**, including:

- Engine type
- Specific fuel consumption (SFC)
- Maximum thrust per engine (jet) or maximum power per engine (prop)
- Dry weight per engine
- Engine bounding box dimension

If you already have them, you can submit the **updated three-view drawings** showing control surfaces, high-lift devices, and the landing gears.

Design Refinements Recall that in previous milestones, your aircraft weight is only an estimation using empirical formulae. Now that you have designed all the components of your aircraft, **update your weight estimates** using the component-weight method. This weight will most likely be different

from your initial estimation. Use the finalized weight and engine option to **calculate the Time to Climb (TTC) again**.

Hint: Aircraft design is an iterative process. In this project, you may find that it is impossible to design an aircraft just in one go. Sometimes you may find that if one requirement is satisfied, the other one cannot. After several design iterations, your aircraft should satisfy all DRO listed in the tender document.

Cost Analysis [Optional] Calculate the corresponding **Direct Operating Cost (DOC)** estimation for your aircraft. You will see which design components affect aircraft's operating cost; discuss it! Analyze the **break even point (BEP)** of your aircraft.

As a **final check**, go back to your Table 2 and see whether your final design (after several iterations) meets the requirements. Also, how does your aircraft compare with other existing similar aircraft in the market and also R-Jet?

3 Submission of Final Report and Presentation

Finally, after months of planing and hard work, you will submit your report and oral presentation (critical design review or CDR) to Ms. Y. Again, take this as the minimum requirement, feel free to go above and beyond and deliver something you can be proud of (you may get some **bonus points!**). Give Ms. Y a professional report to showcase your ability and the aircraft of your design.

3.1 Final Project Report (Due on May 15, 2026)

The report needs to follow the general guidelines/specifications as follows:

- Submit the report as **one PDF file**.
- **Font style: Times New Roman, 11-pt font size and 1.5-spacing.**
- Must be written in good, clear, and concise English.
- **It does not need to be long but complete.**
- Short and simple sentences are the best; just go straight to the point.
- It must include and detail each step of the design phases and analyses.
- **All design rationales and justifications (i.e., why do you make those design decisions) should be explained clearly.**
- For formulas: make sure all variables, notations, and parameters are stated clearly.
- When presenting some graphs/plots: all the axes, colour map, legends must be clear. Must be accompanied with some explanations.

For an **effective, collaborative report writing**, you are highly encouraged to use [L^AT_EX typesetting](#) with [Overleaf](#). L^AT_EX will help you prepare a high-quality report presentation that is suitable for scientific/technical report. It takes a bit of learning curve for first timers, but the efforts will be worth it and you will save a lot of time along the way. It will be highly beneficial and useful for your future professional development too. The teaching team will be happy to assist you in getting familiar with them, if needed. In the past, groups that used L^AT_EX to prepare their report presented a much better report than others (and hence, they got higher grades as well).

If you follow all the milestones, you will have no problems completing the final report and presentation on time. The report should contain:

1. **Cover page** – team name, logo, members, and a perspective view of the designed aircraft
2. **Table of content**
3. **Project’s summary** – refer to the checklist
4. **Main content** – refer to the checklist

Please adhere to project guidelines and fulfill report requirements. Make sure you tick all the boxes in the checklist – if you can’t complete some items, explain why. Failing to do so might make you lose some marks.

3.2 Individual report submission

In addition to submitting the final project report, each student will also submit an individual report separately. There will be a separate submission portal on Canvas, and it is due on the same day (15 May 2026).

From the individual report, we would like to see how much each student knows and understand what’s going on with the project. Hence, if you are fully involved with the project from the beginning, this part would be easy. The individual part should include the following:

1. Statement of contribution (**10 points**)
 - Which analyses and calculations did you do for the project?
 - How much programming did you do?
 - How much report writing did you contribute to?
 - Are there any significant contributions/achievements from your part?
2. Tradeoff analysis (**40 points**)
 - Provide one example or case study of design tradeoff that you observe in your project.
 - What disciplines are involved in this tradeoff study?
 - Perform a simple tradeoff analysis using the aircraft that your group has designed (using **AeroFuse** will make this task very simple, as the interdisciplinary relations are well taken care of and already automatically modeled—at least within the scope of this course). You can simply start by changing one design parameter to improve one discipline, and observe how it affects the other discipline (typically negatively).

Hint: As an aircraft designer, think about a scenario where the tradeoff analysis will be useful to address changes in demands and/or for future growth.

 - Discuss briefly: why is tradeoff analysis important in the aircraft design process?
3. Optimization problem formulation (**30 points**)
 - Define the problem: Clearly explain the current limitations and challenges that need to be overcome from your previous study case. Identify how can optimization help you to find the best solution for your problem.

Hint: Start by performing a short literature review on aircraft design optimization/refinements. Consider the important measure of merit that can benefit from optimization.

- Formulate the optimization problem: Define the objective function, constraints, and decision variables for your optimization problem. Explain how each of these components is related to the problem you have identified.

Note that you do NOT need to perform the optimization as you may not be fully equipped to do so.

4. Self-reflection on the overall learning experience (you may include difficulties, challenges, any profound/interesting/epiphanic experience from the project, etc) **(20 points)**
 - Anything that is different from your idea of “aircraft design” prior to taking this course?
 - What is/are the most interesting/inspiring/insightful thing(s) you learn from this course?

Feel free to discuss with the TAs, but make sure you don't do it last minute!

3.3 Presentation

To help your customer see the awesomeness of your design, please prepare a presentation to showcase what your designed aircraft can offer. Use this opportunity to convince her to choose your aircraft.

- **Slides (for ALL groups) are due on April 29, 2026 and must be submitted to Canvas.**
- **Final presentations will be on April 29, May 6, and May 8, 2026 (3 sessions).**
- Your presentation should fit 20 minutes – time is precious so use it well.
- Focus on general methods, rationales, main results, analyses, conclusions and future works. There is no need to show too many equations (especially those that can be found in the lecture notes). Save the technical details to the final report.
- Aircraft design is not simple so Ms. Y (and other competitors) will definitely have some questions about your design and deliverables. There will be around 5 minutes for Q&A. This is an important part of your grading – both for asking and answering questions.
- All team members must contribute to the oral presentation and Q&A session.
- Dress Code: Please ensure you are dressed professionally in business casual or formal attire.

4 Group Project grading schemes

Even though students will work in a group, each student will obtain a final individual score for the project. The individual score IS will depend on:

1. Group score GS (one score per group, from midterm review, final project presentation and project report);
2. Peer assessment score PA (0 – 100), i.e., how much you contribute to the project, according to your group mates; and

The scoring can be expressed as:

$$IS = GS(0.7 + 0.3 * PA) \tag{1}$$

The peer assessment score corresponds to 30% of the individual's project score¹. Where:

IS : Individual final project score

GS : Group score

PA : Peer assessment score

¹That is, you will obtain the full mark from the project score if you have 100% contribution to the project. If you don't do anything and your final peer assessment score is 0, you will only get 70% of the group score, which is already considered very generous.

5 Software and Tools

5.1 AeroFuse

We have developed a tool for you to use for this course. With it, you should not need many external tools to carry out the project. Many basic usages of **AeroFuse** have been introduced during tutorials and you can explore more of its features thanks to its easy-to-use interface. You will find aircraft design much easier if you realize the power of this platform. The platform's documentation is available at <https://hkust-octad-lab.github.io/AeroFuse.jl/dev/>.

5.2 Airfoil Database

In case you need external resources for reference, some publicly available databases include [Airfoil Tools](#) and [UIUC Airfoil Coordinates Database](#).

5.3 CAD Software

These tools are essential for Mechanical and Aerospace Engineering students and will be used to plot the 3D views.

- **SolidWorks**: Available at Computer Barn
- **AutoDesk Inventor / Fusion 360**: Free for students
- **OpenVSP**: An open-source parametric aircraft geometry tool [\[link\]](#)